CLASSICAL ATOMS & BONDING

General Chemistry I, Lecture Series 3

Qixi Mi

Reading:
OGB8 §§1.4, 3.10

© Qixi Mi 2014–25, except pictures from the Internet.

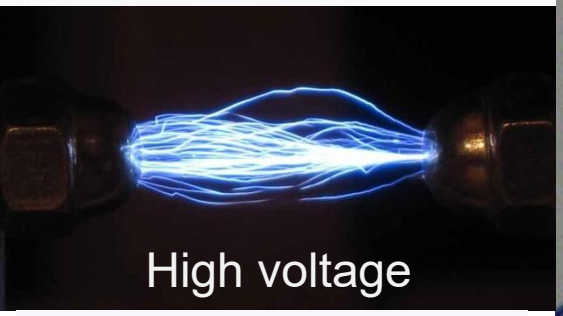
Outline

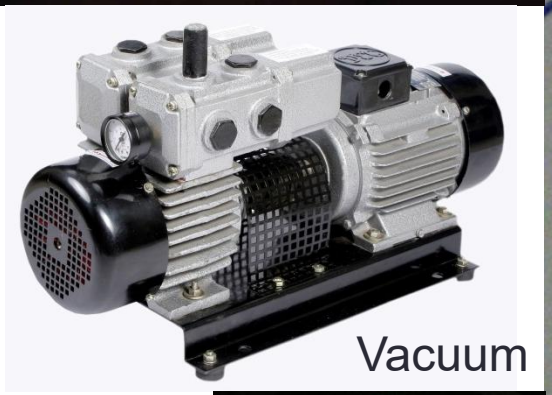
- Atomic structure
  - Electron
  - The Pudding Model
  - The Rutherford Model
  - Neutron

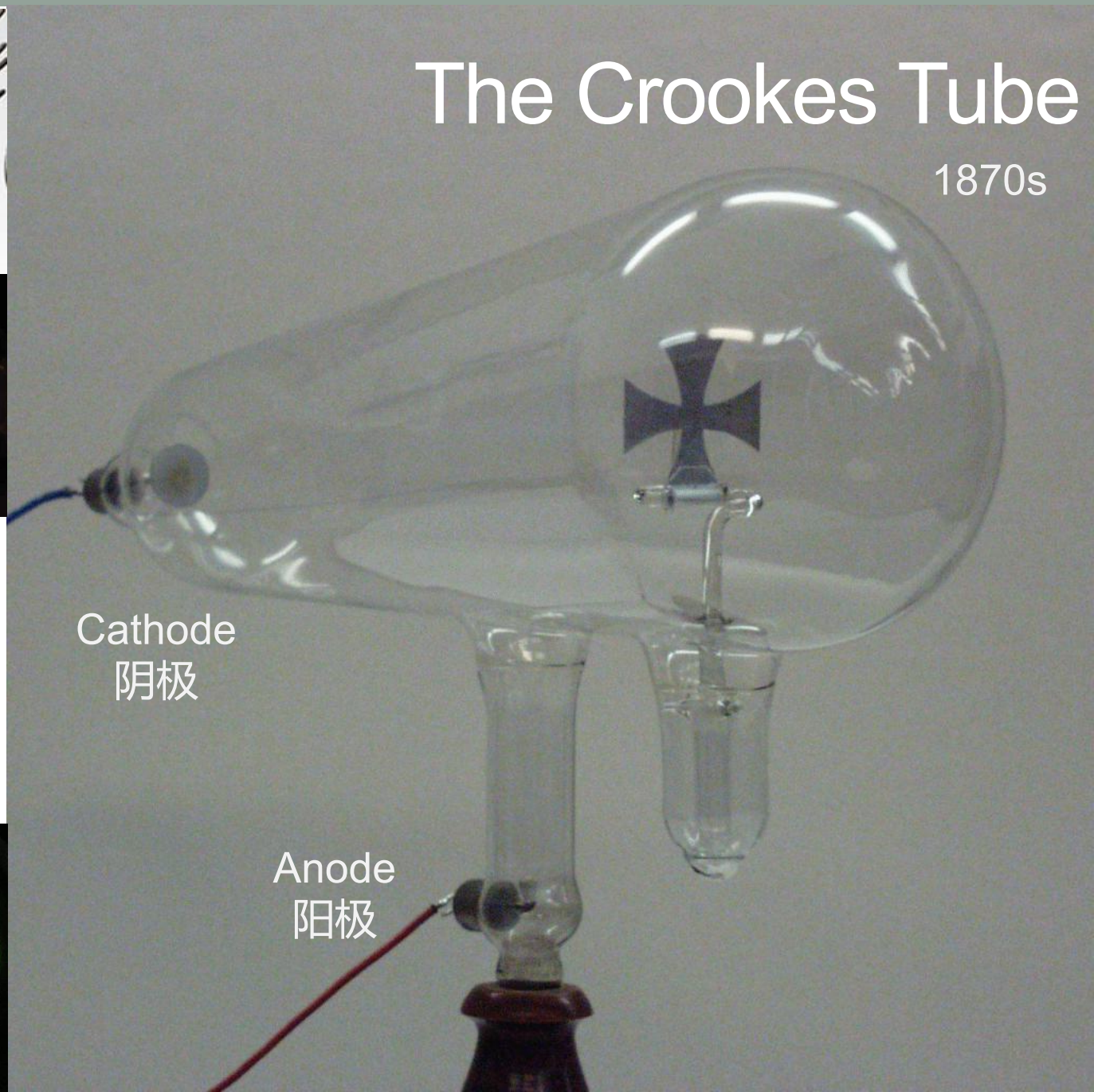
- Lewis diagram for molecules
  - Octet rule
  - Formal charge
  - Lewis acid/base

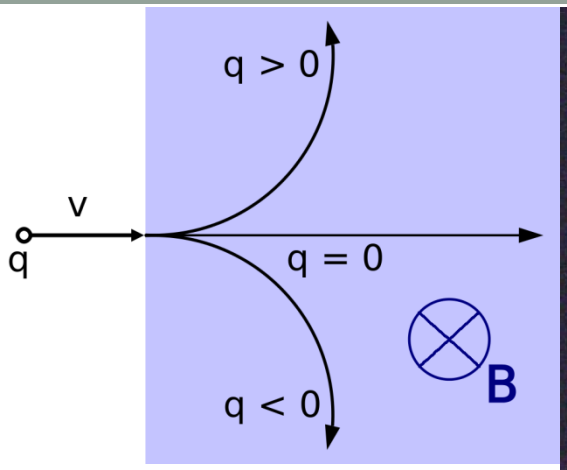
The Crookes Tube

1870s



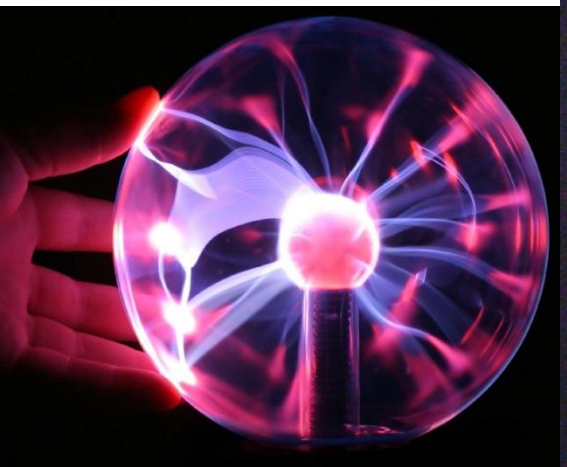


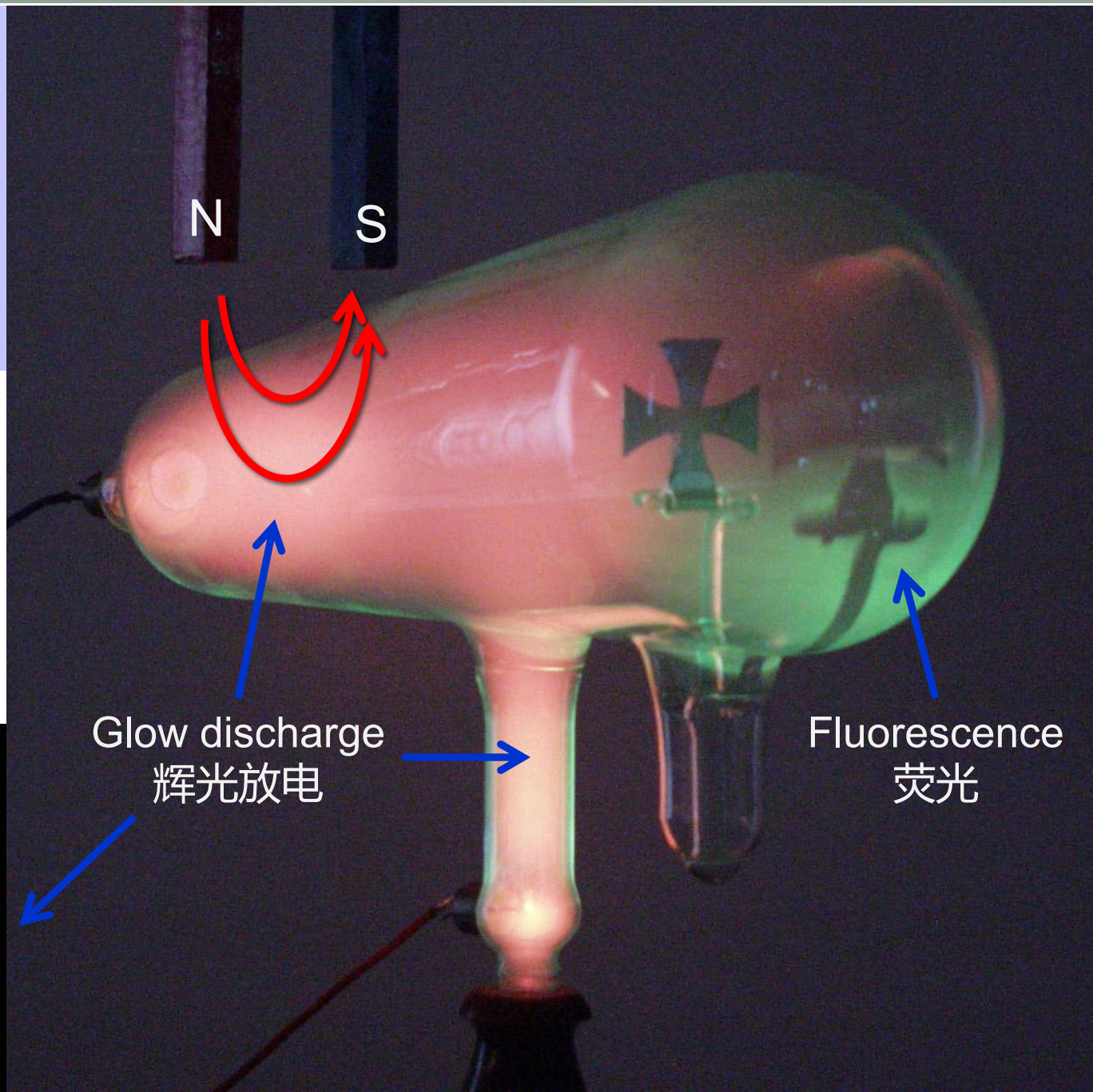




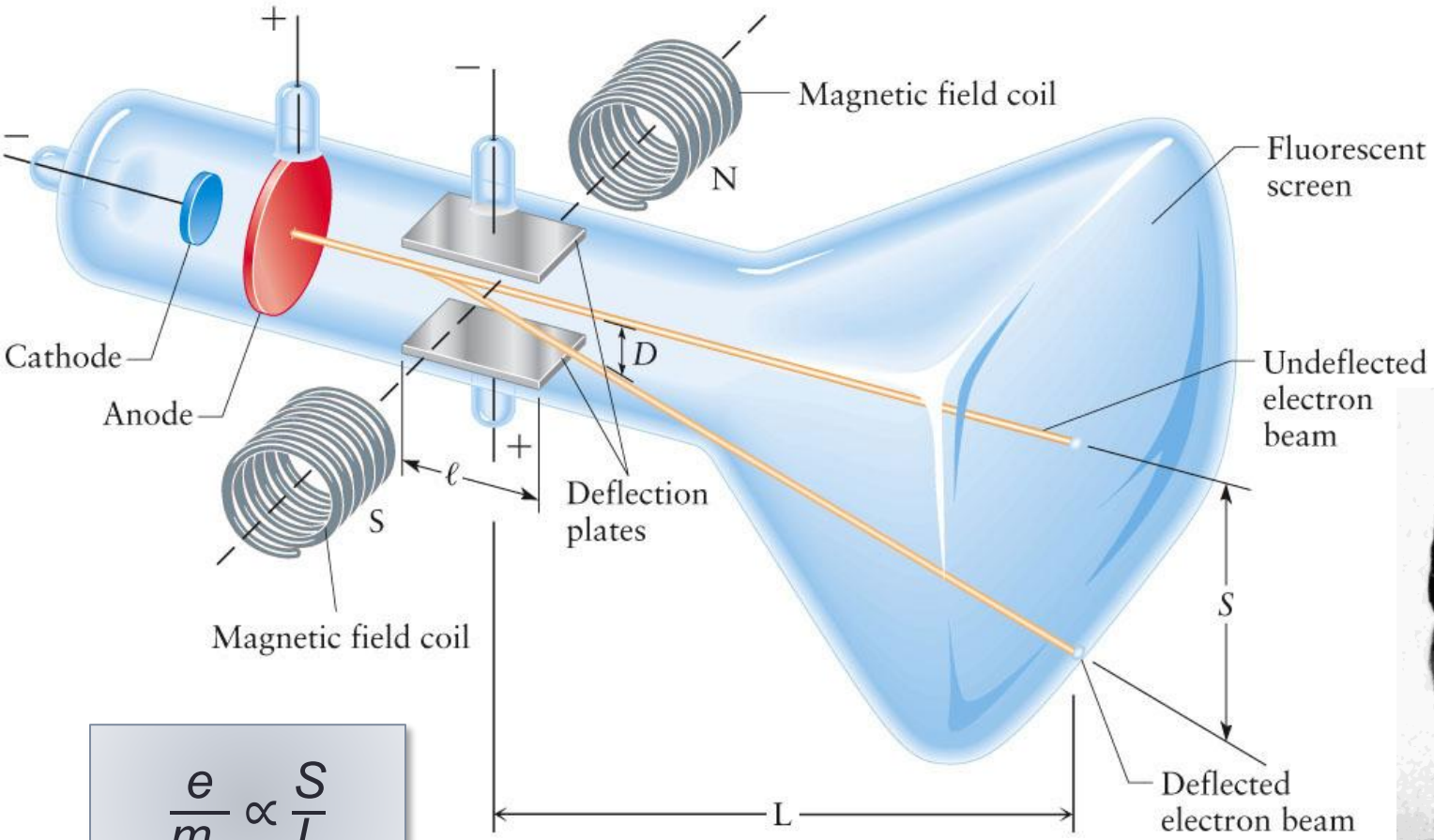
Lorentz force

Particles in the cathode ray are negatively charged.





The Cathode Ray Tube (CRT)

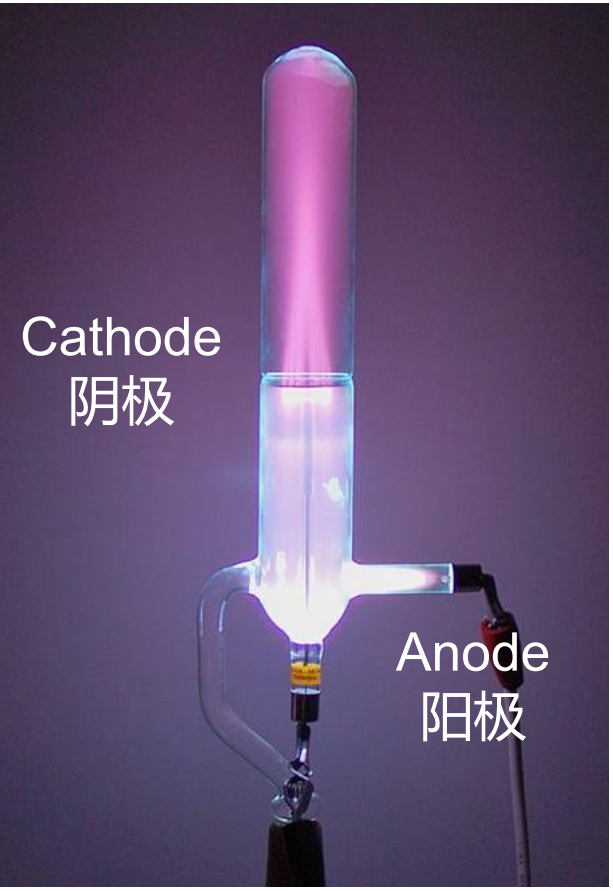


$$\frac{e}{m_e} \propto \frac{S}{L}$$

J. J. Thomson (Cambridge, 1856–1940)

$\frac{e}{m_e}$ is independent of the cathode material or the gas.

Canal Rays from the Anode



- Canal rays are positively charged.
- Canal rays show different charge-to-mass ratios.
- The largest *e*/*m* for anode rays is that of proton

  Q/m for electron = -1.8×10^{11} C·kg^{-1}
  Q/m for proton = $+1.0\times10^{8}$ C·kg^{-1}

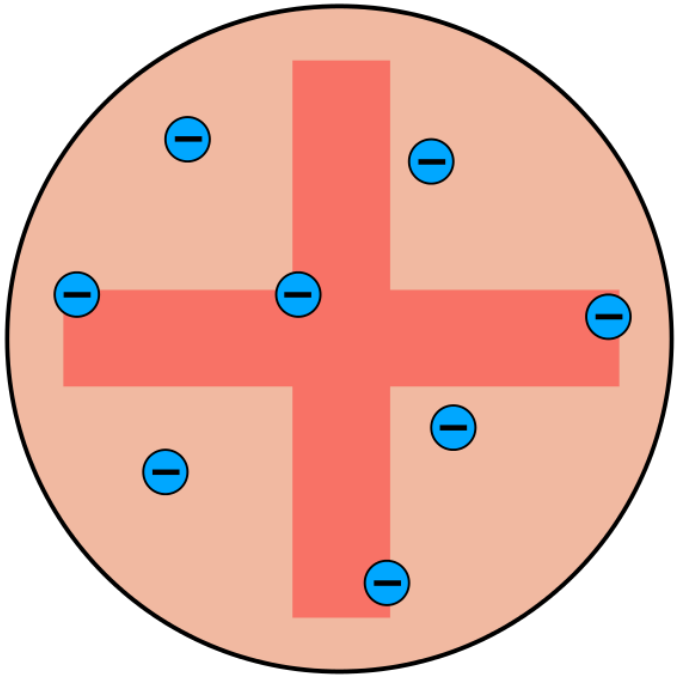
- More experiments showed $Q_p = -e$, so

$$\frac{m_p}{m_e} = 1.8\times10^{3}$$

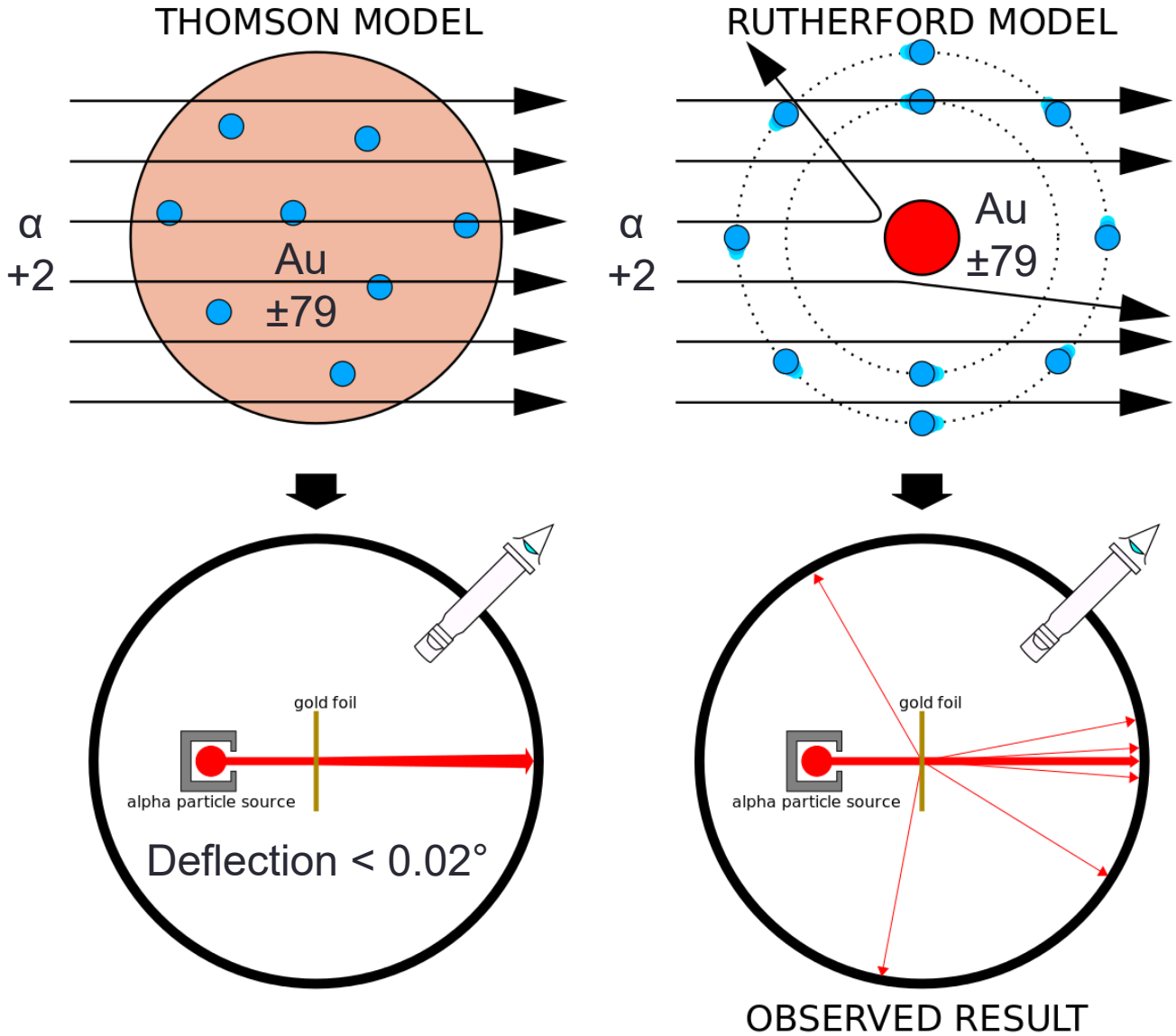
The Plum Pudding Model

J. J. Thomson

- 1897 Discovered electron
- 1904 Proposed the pudding model
- 1906 Nobel Prize in Physics



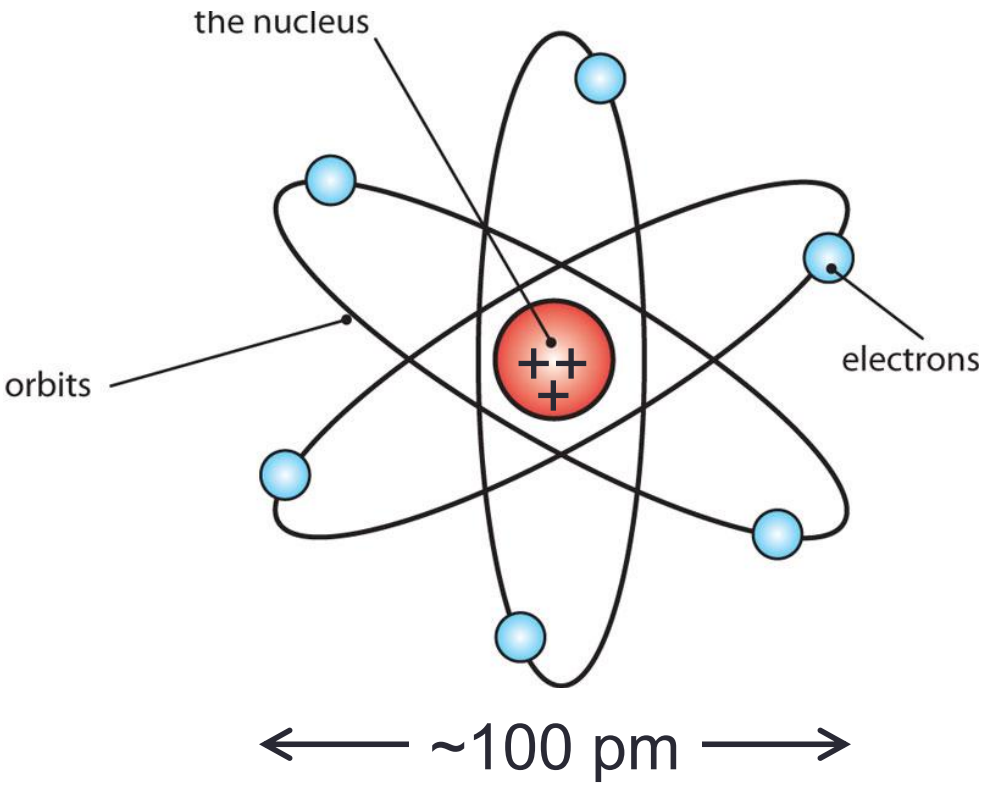
Rutherford's Gold Foil Experiment (1911)



Ernest Rutherford (Cambridge, 1871–1937)

The Rutherford Model

< 0.01 pm

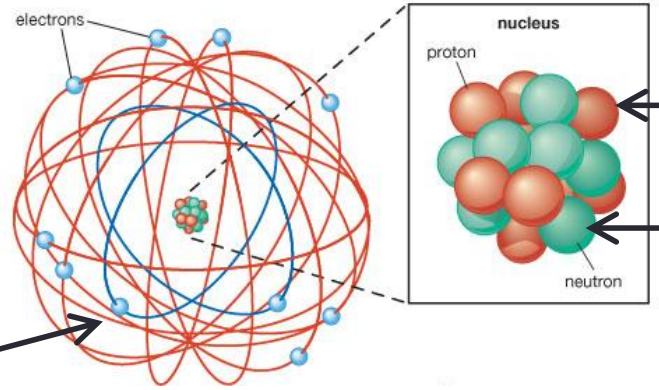


IAEA
International Atomic Energy Agency

The Au nucleus has 79 positive charges but weighs 197 mass units?

Discovery of Neutron

1920 Rutherford proposed a proton-electron model for the nucleus

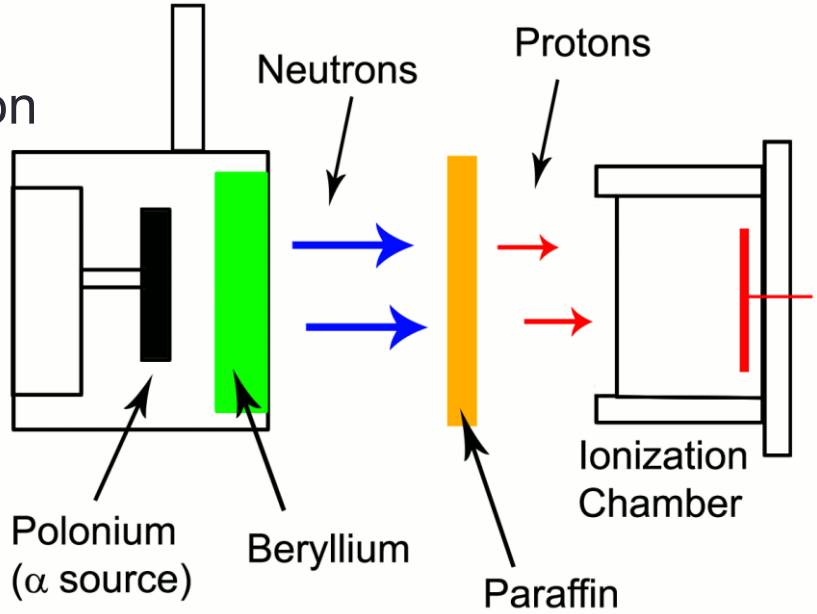


79 protons

118 neutrons

79 electrons

1932 Chadwick discovered the neutron



James Chadwick (Cambridge, 1891–1974)

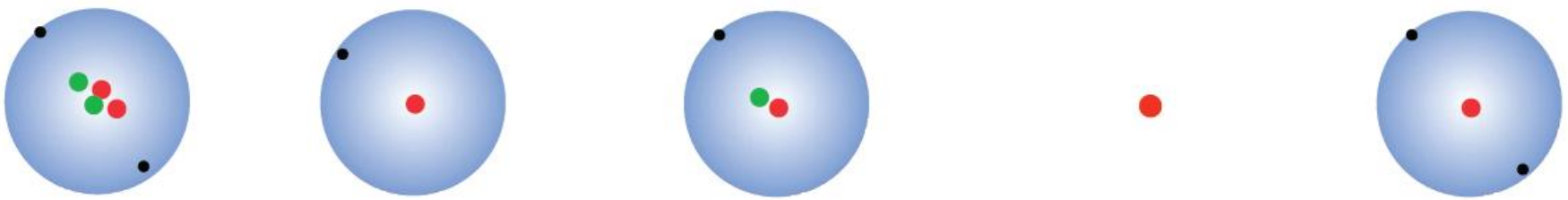
Summary

Mass number
$A = p + n$

Atomic number
(charge number)
$Z = p$

$^{A}_{Z}X$

Anode 阳极
Anion 阴离子
Cathode 阴极
Cation 阳离子



Helium, $^{4}_{2}He$

Protium, $^{1}_{1}H$

Deuterium, $^{2}_{1}H$

Hydrogen ion (proton), H^{+} Cation

Hydride, H^{-} Anion

Hydrogen, $_{1}H$

Ion

Generalized hydrogen element 元素

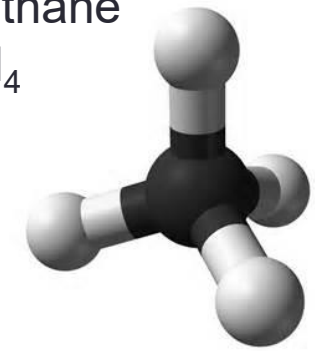
Outline

- Atomic structure
  - Electron
  - The Pudding Model
  - The Rutherford Model
  - Neutron

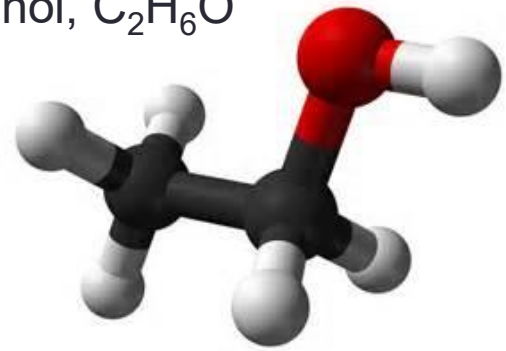
- Lewis diagram for molecules
  - Octet rule
  - Formal charge

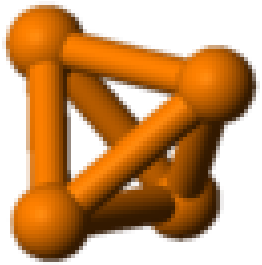
How Are Atoms Bonded into Molecules?

Methane
CH_4

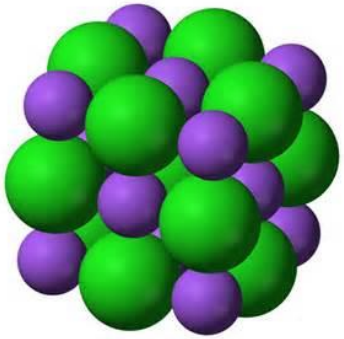


Ethanol, C_2H_6O

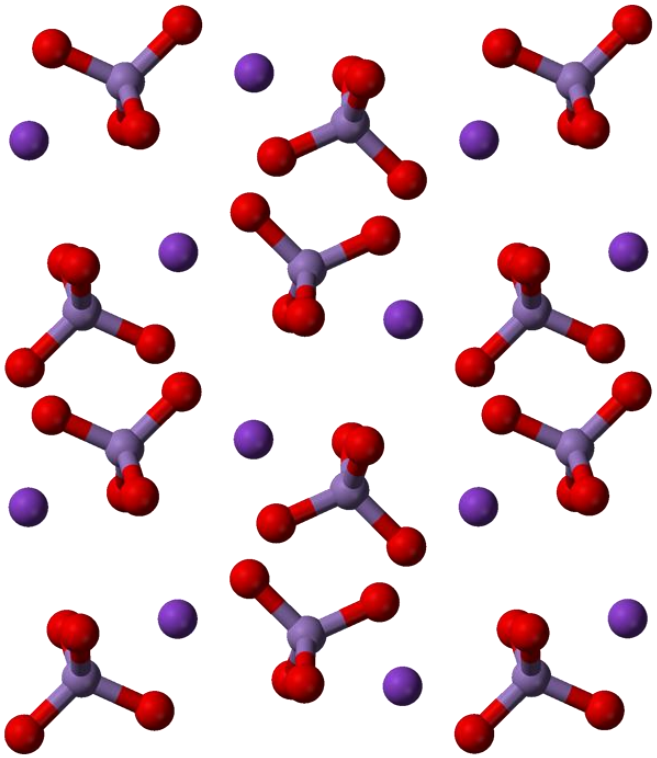




White phosphorus, P_4



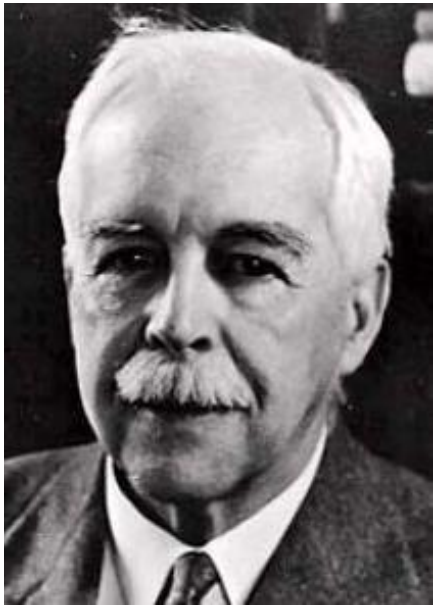
Sodium chloride, NaCl



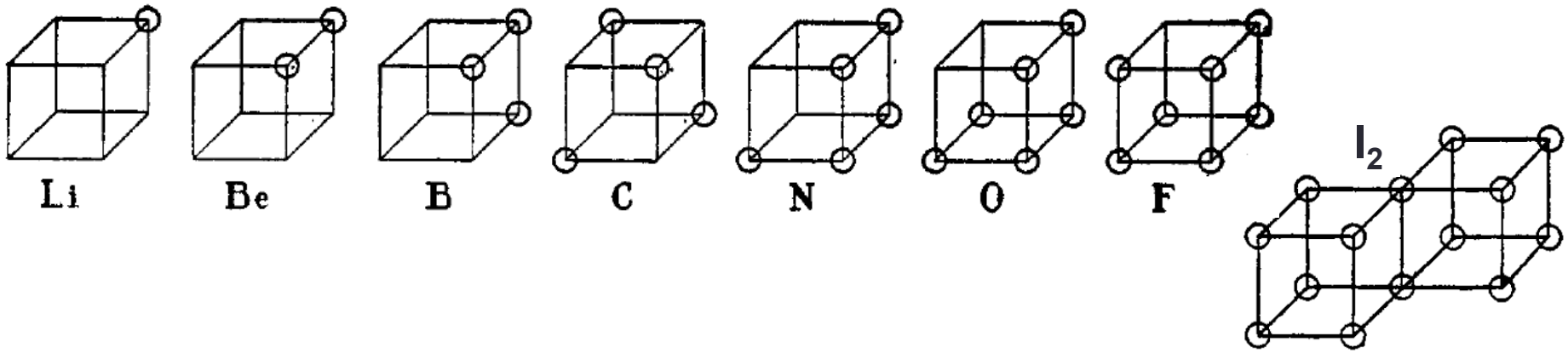
Potassium permanganate, $KMnO_4$

Lewis Diagram: Rules

1. A shared pair = a covalent bond.
   A lone pair = no bonding.
2. Each atom achieves its own noble-gas shell of electrons.
3. Share as many electrons as possible.



Gilbert N. Lewis
(Berkeley, 1875–1946)

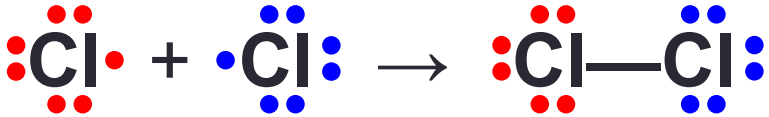


Lewis, G. N. "The Atom and the Molecule", *J. Am. Chem. Soc.* **1916**, *38*, 762–785.

Lewis Diagram: Examples

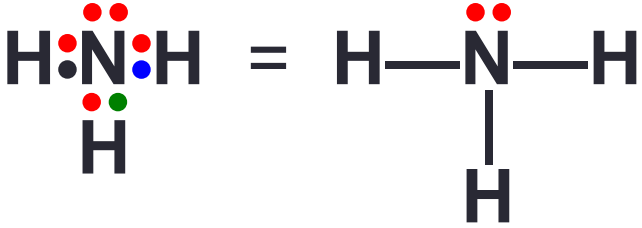
$1s^2\ 2s^2\ 2p^6$

$2s^1$ $2s^2\ 2p^5$



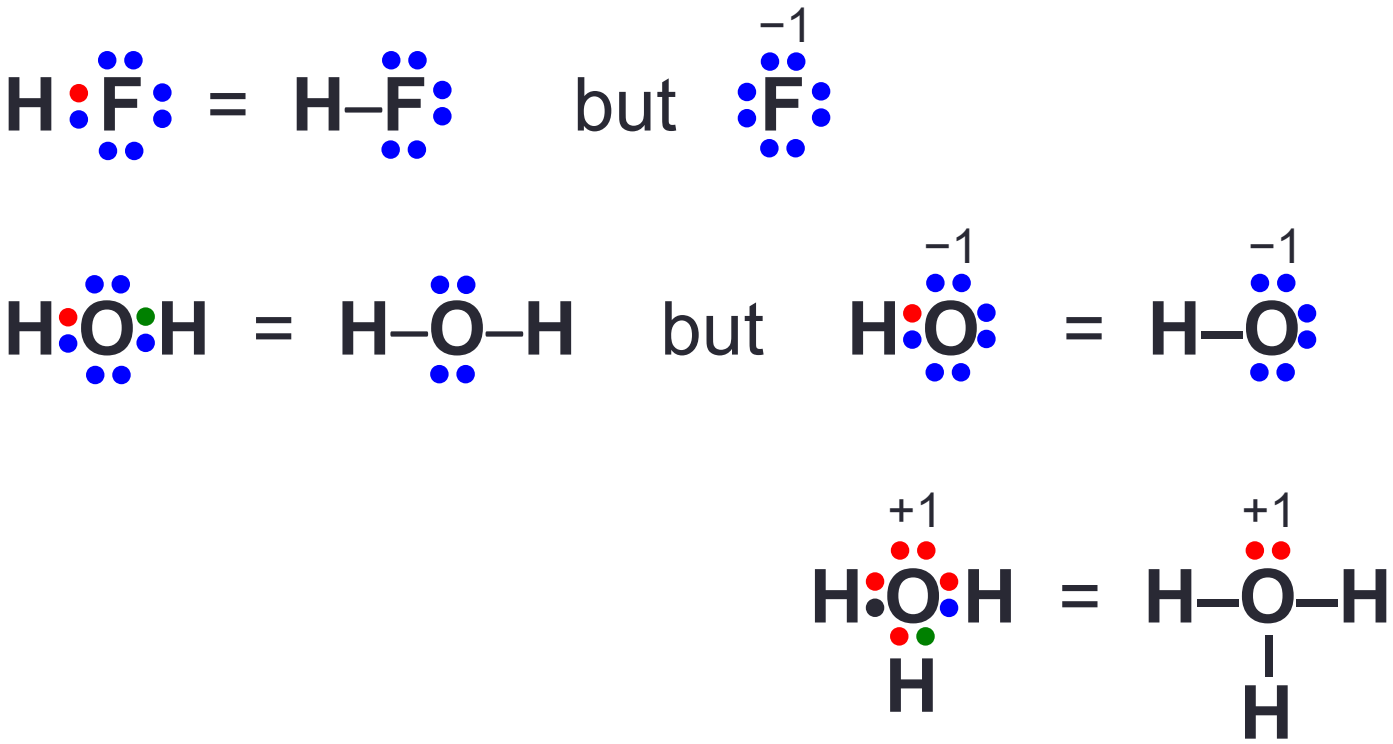
$3s^2\ 3p^5$

$2s^2\ 2p^3$



$2s^2\ 2p^2$

Lewis Diagram: Ions

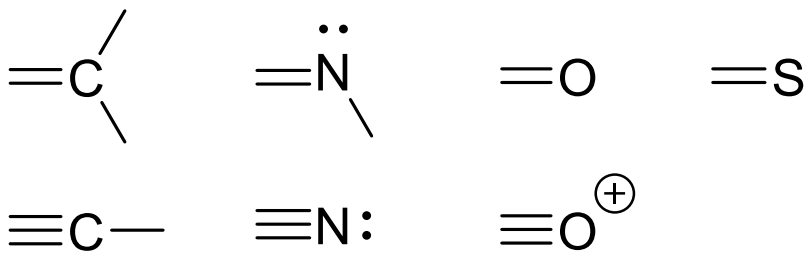


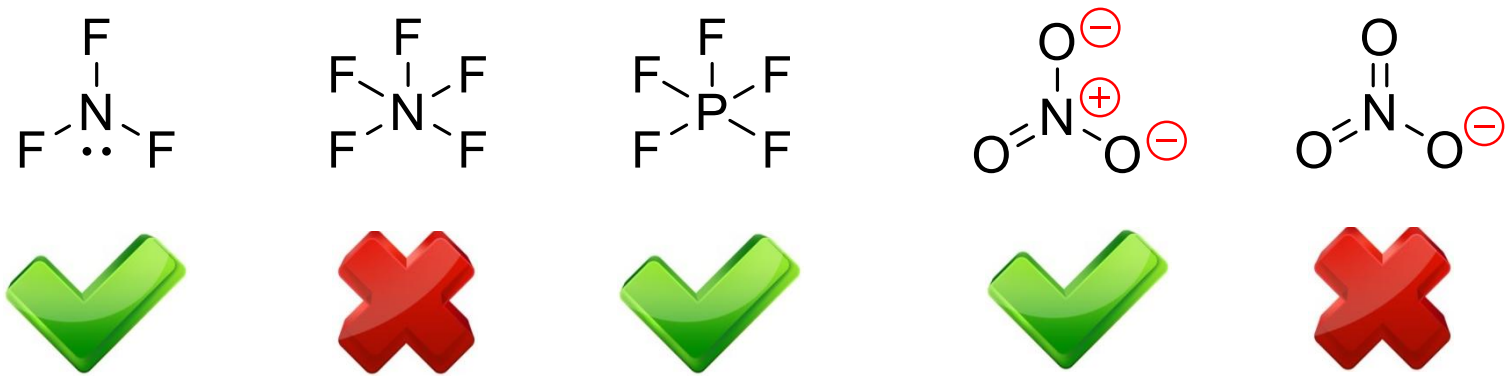
Formal Charge

Formal charge
形式电荷

Number of Valence Electrons

Electrons	0	1	2	3	4	5	6	7	8
Valence	0	1	2	3	4	3	2	1	0
Examples	H^+			C^+	N^+	O^+			
		H		B	C	N	O	F	Ne
					B^-	C^-	N^-	O^-	F^-





Oxidation Numbers: Rules

Rule 1 Oxidation numbers sum up to the total charge.

Rule 2 H, Li, Na, K … = +1; F, Cl, Br … = −1; O = −2.

Rule 3 Elements on the lower left corner prefer a (+) oxidation number;
Elements on the upper right corner prefer a (−) oxidation number.

+1	+2											+3	−4 +4	−3 +5	−2 +6	−1 +7	
H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn

Only one common oxidation number

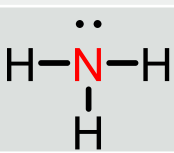
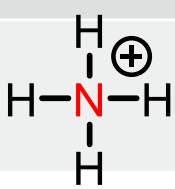
Formal Charge *vs*. Oxidation Number

Formal Charge 形式电荷

= Correction to octet rule

Oxidation number 氧化数

= Number of lost electrons

F^-, Cl^-	FC = −1	ON = −1	FC = ON
: N≡N :	FC = 0	ON = 0	FC = ON
	FC = 0	ON = −3	FC = ON + 3
	FC = 1	ON = −3	FC = ON + 4

$$\sum FC = \sum ON$$

Next lecture series: Bohr Model

Reading: OGB8 §§4.2, 4.3